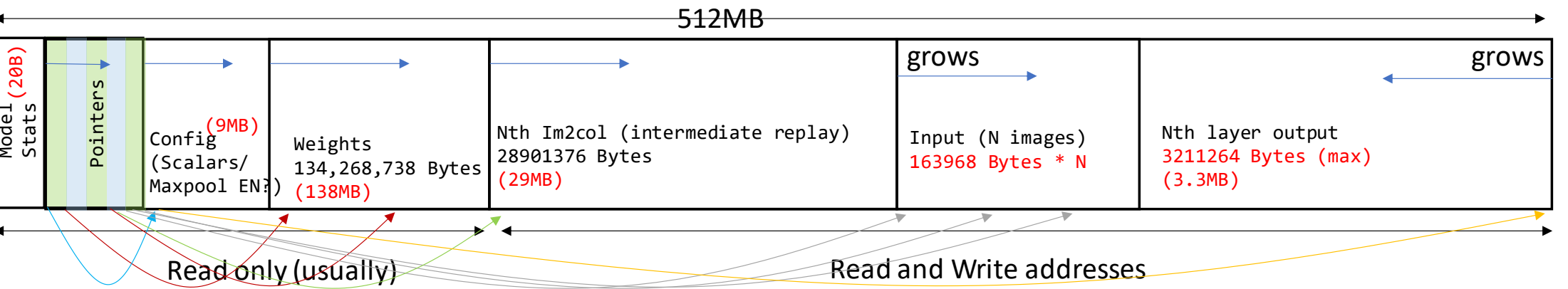


DRAM Memory Map



Model Stats	
Type	Pointer Tuples
Layer Conifgs	24
Weights	16
Im2col	16
input	5
Output	24

Configs and Scalars

	Unique	Layers <24>	Bytes/<setting>	Total Bytes
Conifgs	6	24	64	9126

# Pointers	StartLayer	EndLayer	NextConfigAddr	Bytes
24	?	?	+(16Bytes)	24*12

9126B

288B

Weights

	Channels/layer	Bytes/channel	Total Bytes
Weights#L1	64	?	?
Weights#L2	64	?	?
Weights#L3	128	?	?
Weights#L4	128	?	?
Weights#L5	256	?	?
Weights#L6	256	?	?
Weights#L7	256	?	?
Weights#L8	512	?	?
Weights#L9	512	?	?
Weights#L10	512	?	?
Weights#L11	512	?	?
Weights#L12	512	?	?
Weights#L13	512	?	?
Weights#L14	<dense> 1?	?	?
Weights#L15	<dense> 1?	?	?
Weights#L16	<dense> 1?	?	?

138 MB

# Pointers	Start Layer	End Layer	NextChannelAddr	Bytes
3	?	?	+X1	12
3	?	?	+X2	12
3	?	?	+X3	12
3	?	?	+X4	12
3	?	?	+X5	12
3	?	?	+X6	12
3	?	?	+X7	12
3	?	?	+X8	12
3	?	?	+X9	12
3	?	?	+X10	12
3	?	?	+X11	12
3	?	?	+X12	12
3	?	?	+X13	12
3	?	?	+X14	12
3	?	?	+X15	12
3	?	?	+X16	12

192B

im2col

	Channels/layer	Bytes/channel	Total Bytes
Inputs#L1	3	?	?
Inputs#L2	64	?	?
Inputs#L3	64	?	?
Inputs#L4	128	?	?
Inputs#L5	128	?	?
Inputs#L6	256	?	?
Inputs#L7	256	?	?
Inputs#L8	256	?	?
Inputs#L9	512	?	?
Inputs#L10	512	?	?
Inputs#L11	512	?	?
Inputs#L12	512	?	?
Inputs#L13	512	?	?
Inputs#L14	<dense> 1?	?	?
Inputs#L15	<dense> 1?	?	?
Inputs#L16	<dense> 1?	?	?

# Pointers	Start Layer	End Layer	NextChannelAddr	Bytes
3	?	?	+X1	12
3	?	?	+X2	12
3	?	?	+X3	12
3	?	?	+X4	12
3	?	?	+X5	12
3	?	?	+X6	12
3	?	?	+X7	12
3	?	?	+X8	12
3	?	?	+X9	12
3	?	?	+X10	12
3	?	?	+X11	12
3	?	?	+X12	12
3	?	?	+X13	12
3	?	?	+X14	12
3	?	?	+X15	12
3	?	?	+X16	12

Max (Bytes of expanded)
29 MB

192B

Input

	Channels/Image	Bytes/channel	Total Bytes
Inputs#1	3		150528
Inputs#L2	3		150528
Inputs#L3	3		150528
Inputs#L4	3		150528
Inputs#L5	3		150528

~800KB (5images)
~160KB (1image)
169KB * N images

# Pointers	Start Image	End Image	NextChannelAddr	Bytes
3	?	?	+50176 (bytes)	12
3	?	?	+50176 (bytes)	12
3	?	?	+50176 (bytes)	12
3	?	?	+50176 (bytes)	12
3	?	?	+50176 (bytes)	12

60B
12*N images

Output (WIP)

Configurations of each layer: Start and end message for Layer- n

- End of Weights / end of input / end of intermediate output b/w layers
 - End weights (and) End of inputs-replay
 - start next layer -> send a configuration message (we have dedicated bus for message passing or responses from memory controller)
 - start next inference
 - End of the input
 - After first time end of inputs for each layer, for the next time bypass im2col
 - After everytime end of inputs (first time in each layer / replayed inputs) push weights.
- Configuration messages:
 - *Program* quantizers with scalar values (per layer or a fixed value through out the inference)
 - Im2col -> input dimensions
 - Maxpooling -> enable/ disable (after convolution blocks), input dimensions to manage pushing the data into *correct* FIFO.
 - Dropout -> enable/disable (fc blocks)
 - Send starts and ends (required for im2col and weight blocks)

Configuration Packet format (Few example configurations)

Im2col

ID	Org/Replay	lp-matrix width
----	------------	-----------------

Weight Block

ID	#Layer, #Weights	Start/End
----	------------------	-----------

Quantisation Block

ID	Scalar Value	#Shift
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Activation Type

ID	Type: ReLU, GeLU	Saturation: 6,8,256
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Pooling

ID	Max/Avg	EN/DS
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Fc: dropout (only scaling during inference)

ID	EN/DS	#Dropout-Factor
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